

ECSE-4040

Digital Electronics

Lecture 22

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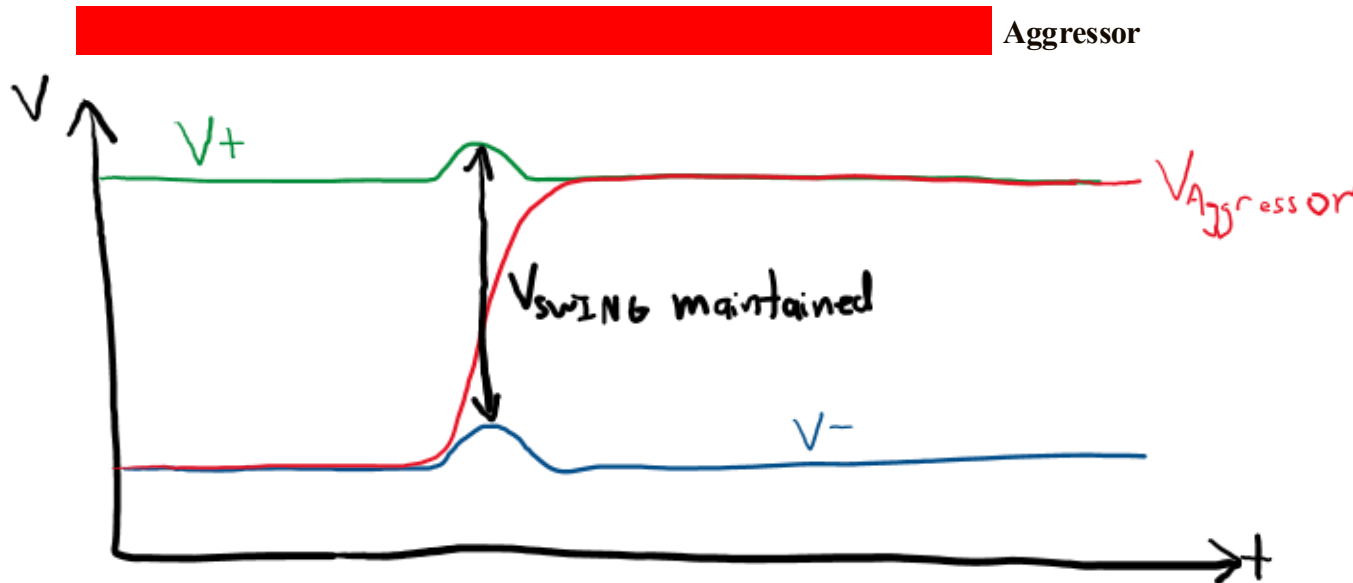
References: Chapter 7, HJS
Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

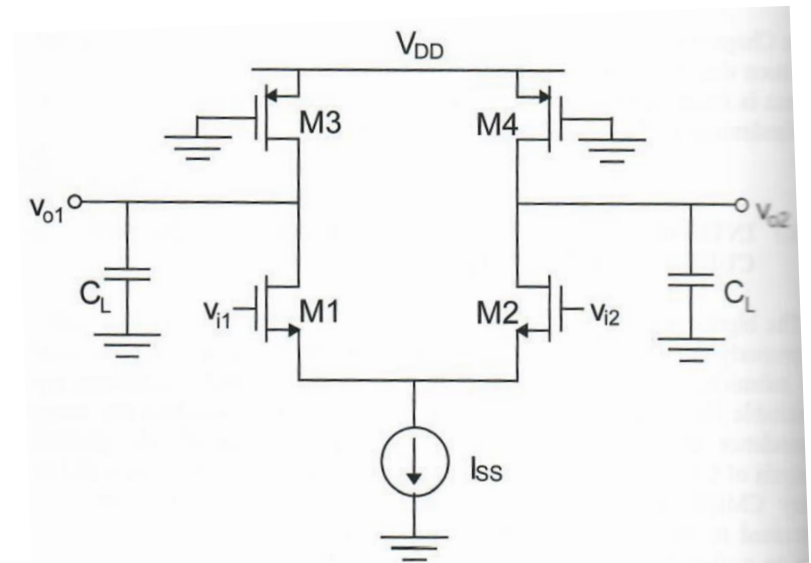
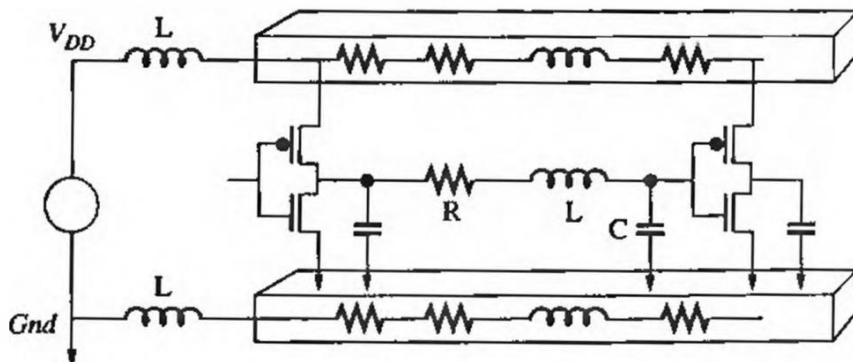
- CML
 - Advantages over CMOS
 - HDMI Driver
- Transmission Gates
 - NMOS, PMOS, CMOS
 - Logic Gates
 - RC Model

Differential Pair



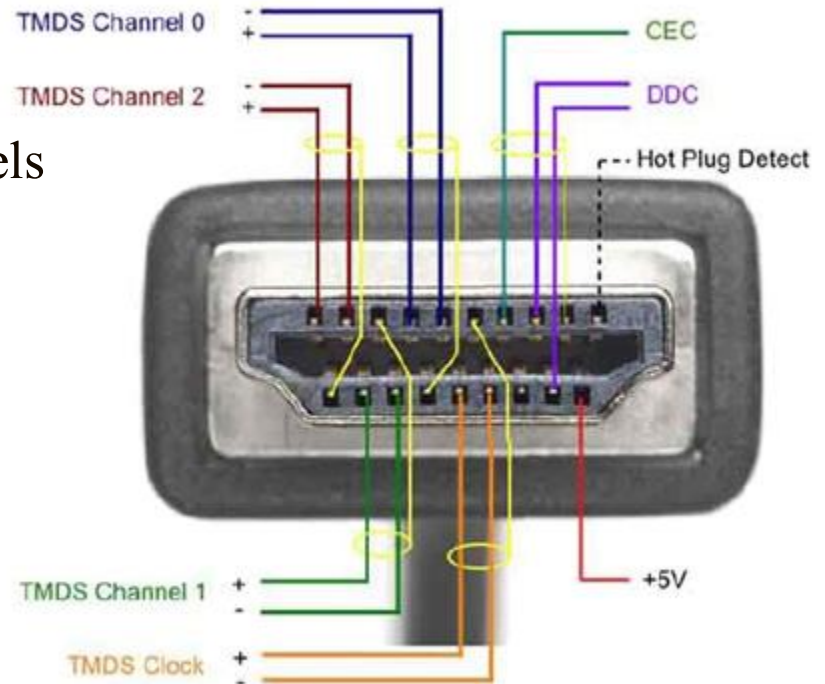
Differential vs CMOS Logic

- CMOS inherently single-ended
- CMOS is internally noisy
 - Strong current spikes
 - Causes unpredictable supply voltage drops, noise.
- Differential logic low internal noise
 - Constant current
 - Constant $I \cdot R$ losses
 - Little $-L \cdot di/dt$ losses



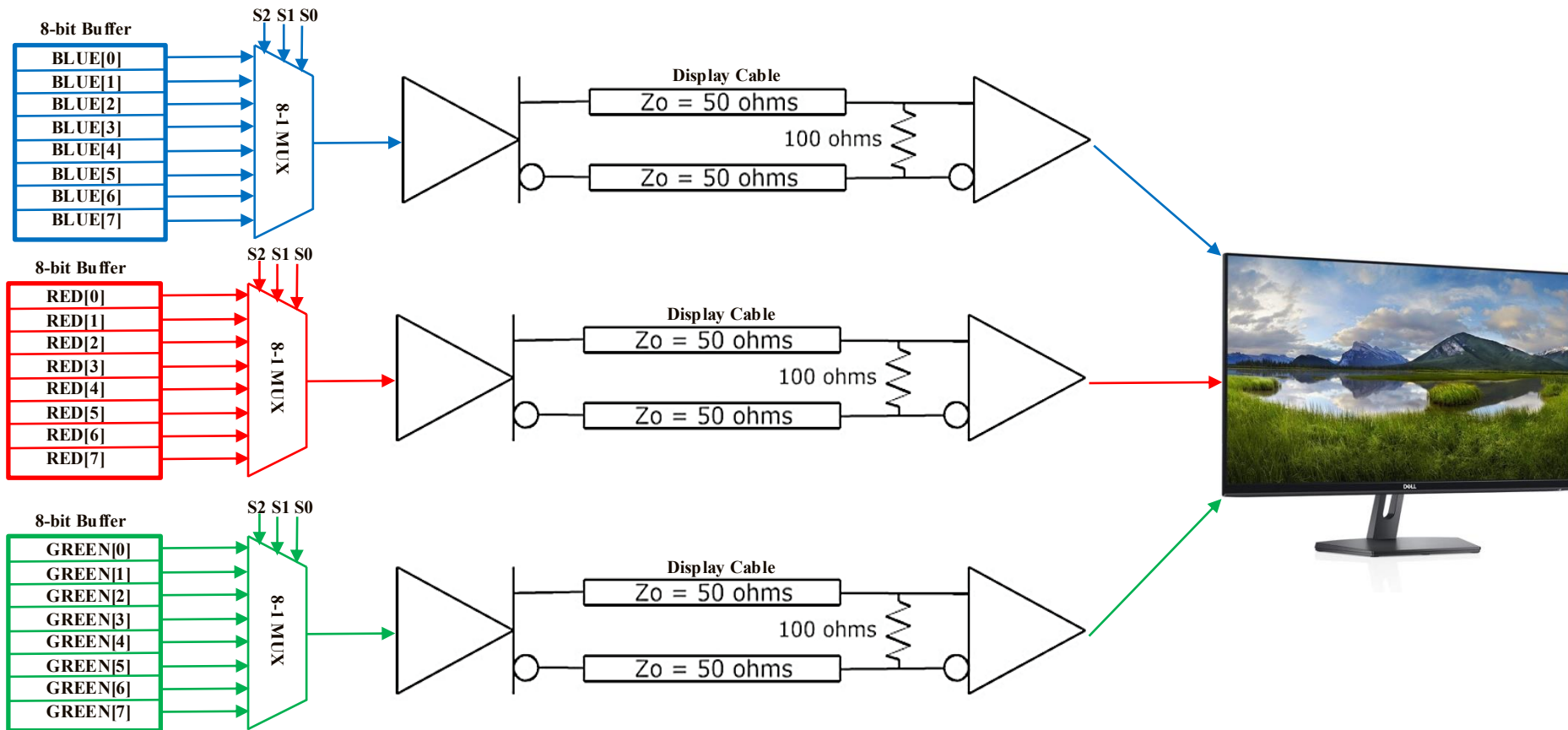
HDMI

- Display:
 - 60 Hz
 - Send full image with all pixels 60 times per second
 - 3 colors
 - Red
 - Blue
 - Green
 - One transmitter per color
 - 8-bits per color
 - If 1920x1080 pixel image:
 - 2,073,600 pixels
 - 16,588,800 bits
 - 995,328,000 bits/s
 - ~ 1GHz data rate required

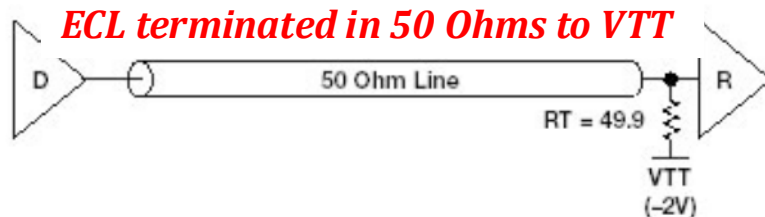


https://support.biamp.com/Tesira/Video/Video_connectors_and_cables

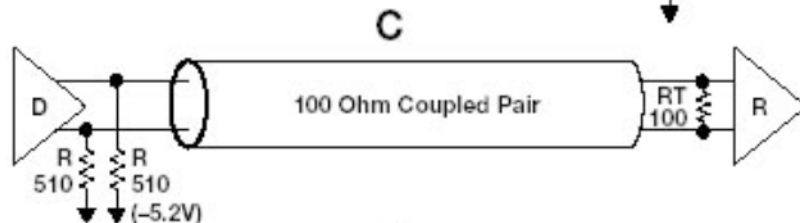
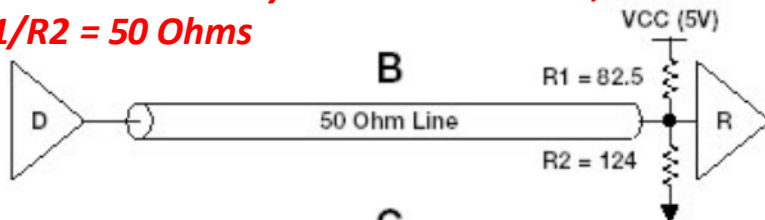
HDMI



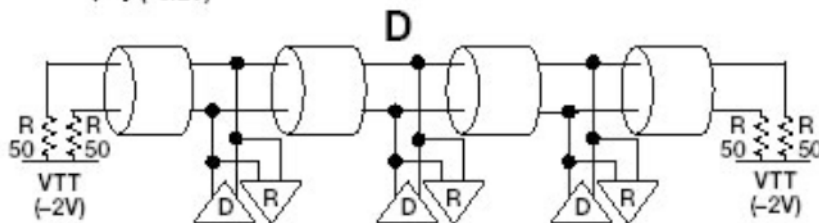
Summary (TI-Bus Interface technical notes)



PECL terminated by Thevenin network, R1/R2 = 50 Ohms



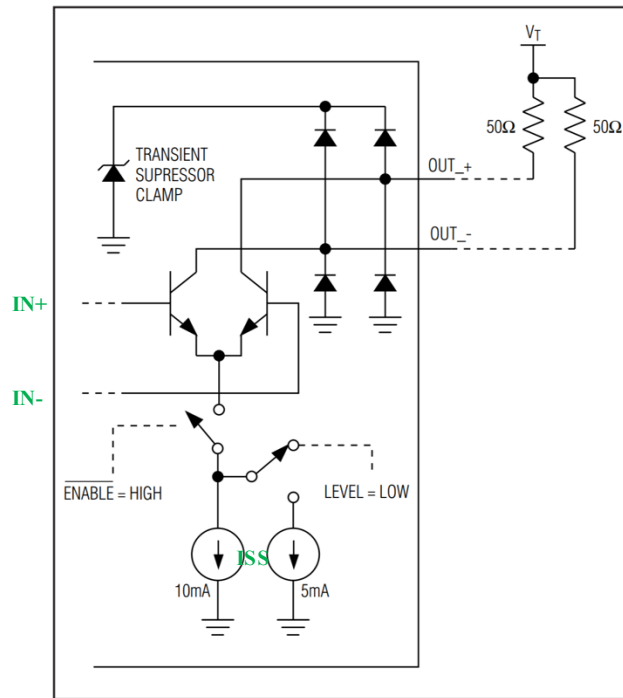
Differential ECL terminated by 100 Ohm parallel termination



Differential ECL Multipoint bus terminated by 50 Ohms to VTT at each end of the bus.

**LVDS, CML, ECL-
differential interfaces with
odd voltages

HDMI Driver



- MAXIM MAX3814 HDMI Driver
- CML
 - Bipolar
- ISS
 - $10mA$ or $10mA + 5mA$
- 50Ω load
- $1.05V V_{SWING}$
- Diodes
 - Static charge protection.

Pass Transistor/Transfer Gate

- On (*Passing 0V or V_{DD}*):

NMOS- $V_{GS} = V_{DD}$

$V_{in} = 0V = V_S \rightarrow V_D = 0V$

PMOS- $V_{SG} = 0V$

$V_{in} = V_{DD} = V_S \rightarrow V_D = V_{DD}$

- On (*Passing V_{DD} or 0V*):

NMOS- $V_{GS} = V_{DD}$

$V_{in} = V_{DD} = V_D \rightarrow V_S = V_{DD} - V_{Tn}$

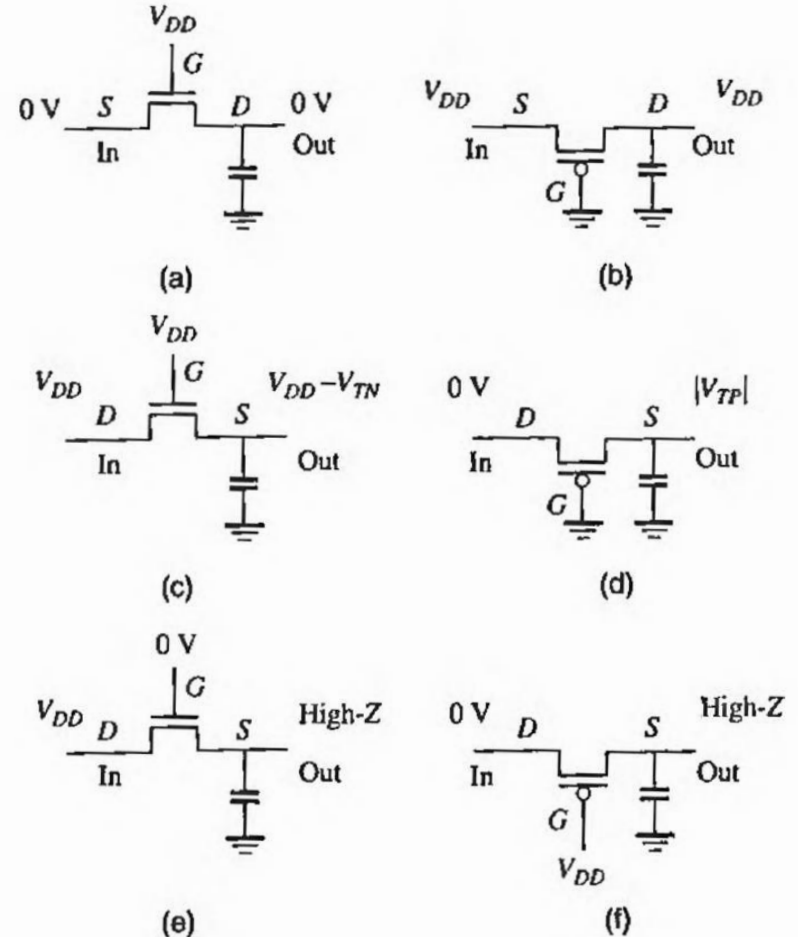
PMOS- $V_{SG} = 0V$

$V_{in} = 0V = V_D \rightarrow V_S = |V_{Tp}|$

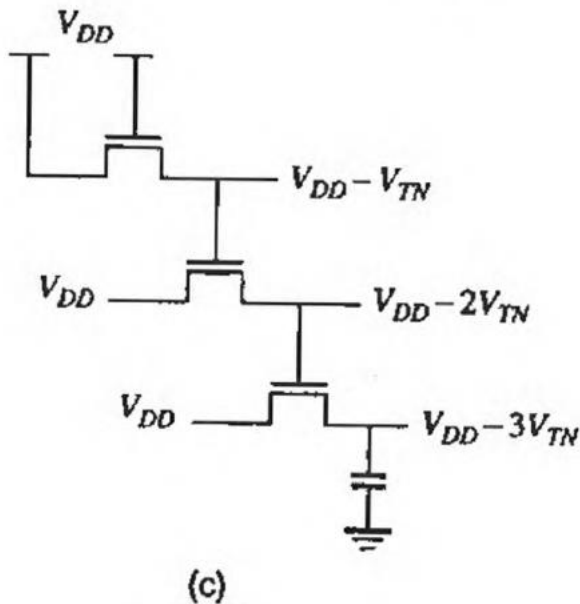
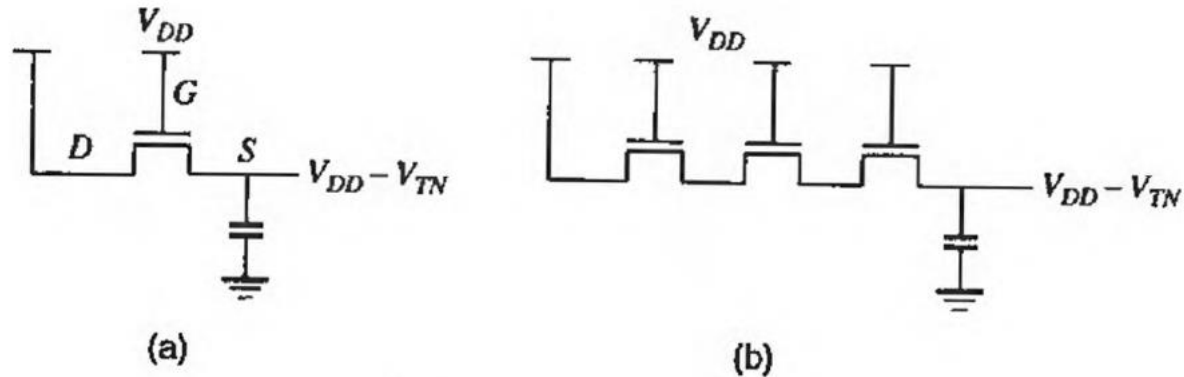
- Off:

NMOS/PMOS- $V_{GS} = 0V/V_{DD}$

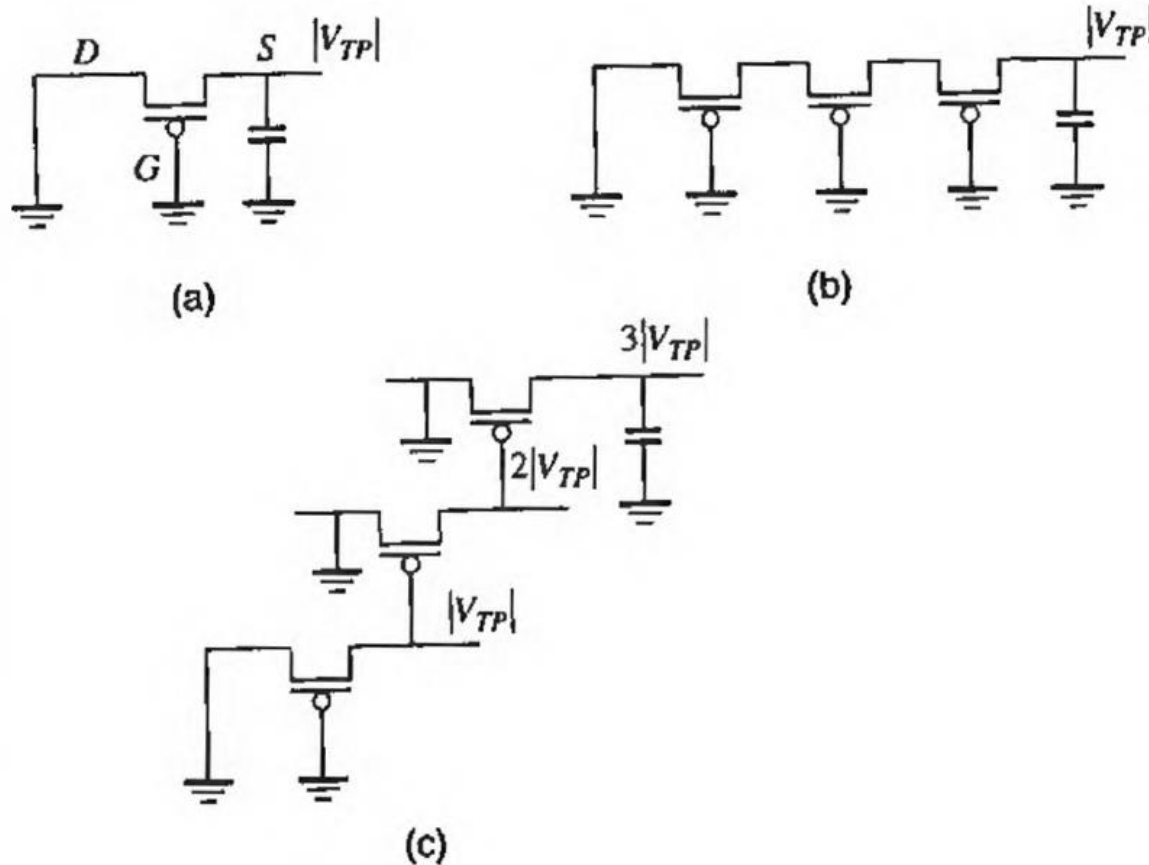
$V_{in} = V_{DD}/0V \rightarrow V_S = \text{High Z}$



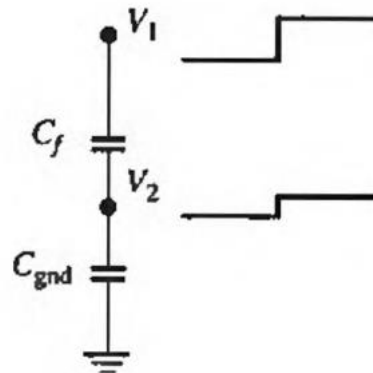
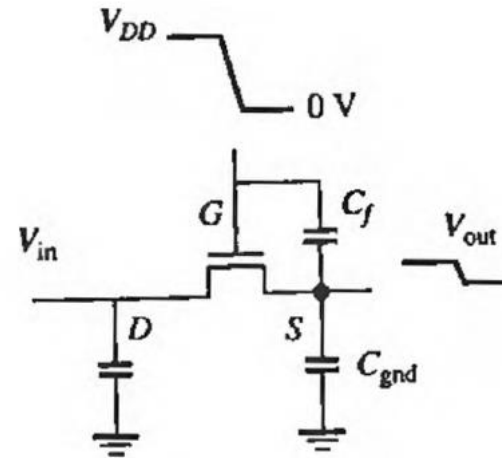
NMOS Pass Gate Configurations



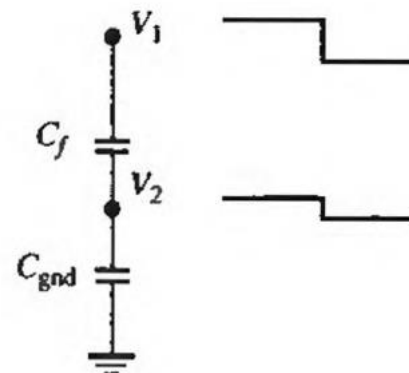
PMOS Pass Gate Configurations



Capacitive Feedthrough



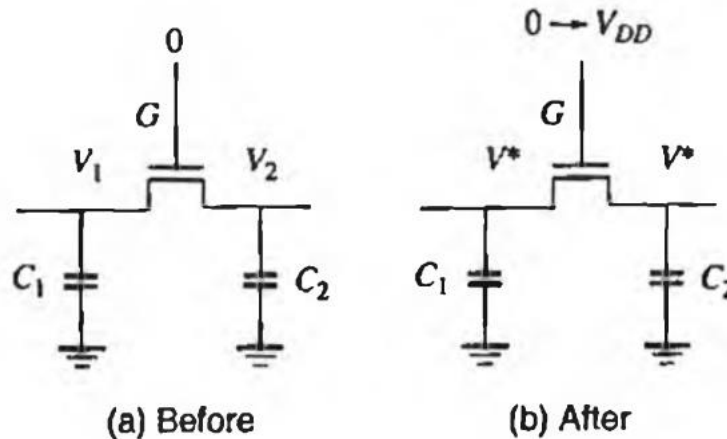
(a) Positive feedthrough



(b) Negative feedthrough

Charge Sharing

$$Q_{\text{total}} = C_1 V_1 + C_2 V_2$$

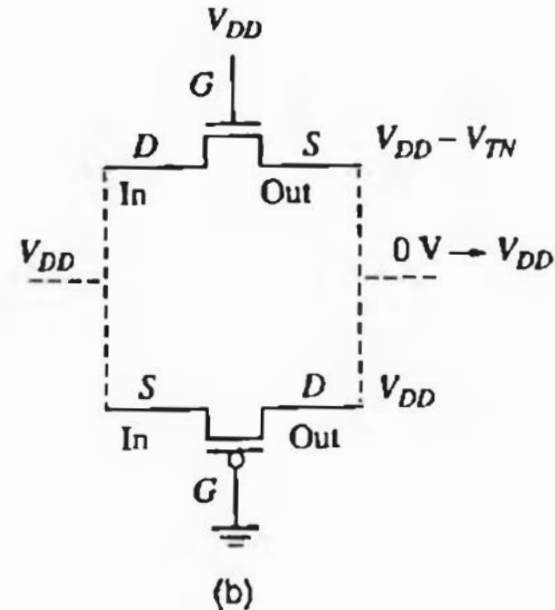
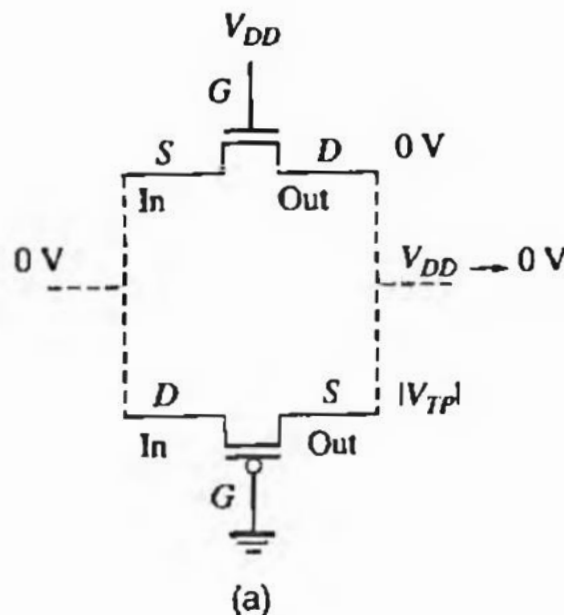
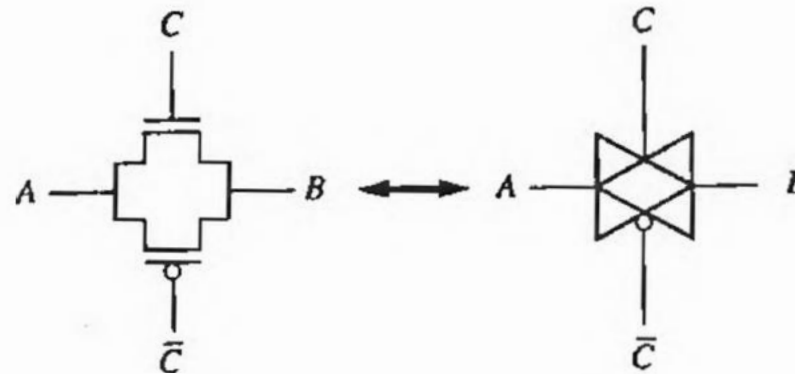


$$Q_{\text{total}} = (C_1 + C_2) V^* \quad V^* = \frac{(C_1 V_1 + C_2 V_2)}{C_1 + C_2}$$

Other Sources of Charge Loss

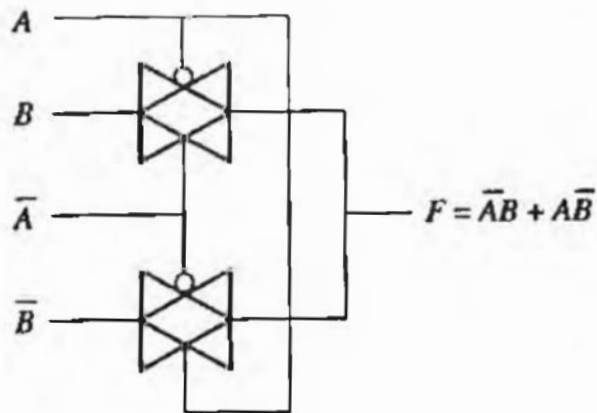
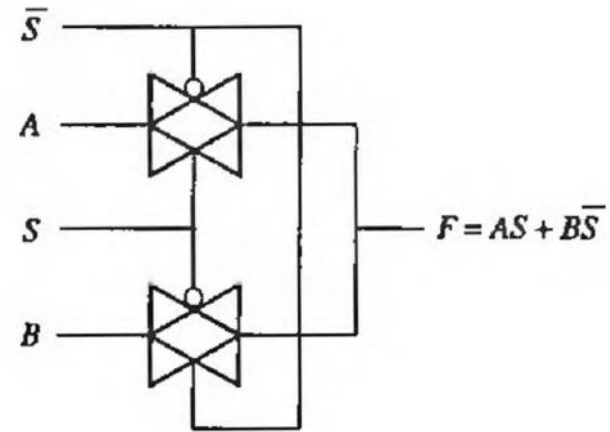
- Reverse-biased leakage current of drain junction as well as subthreshold leakage
- Noise injection from neighboring wires
- Alpha particles due to cosmic rays from space, discharging output node

CMOS Transmission Gate Logic

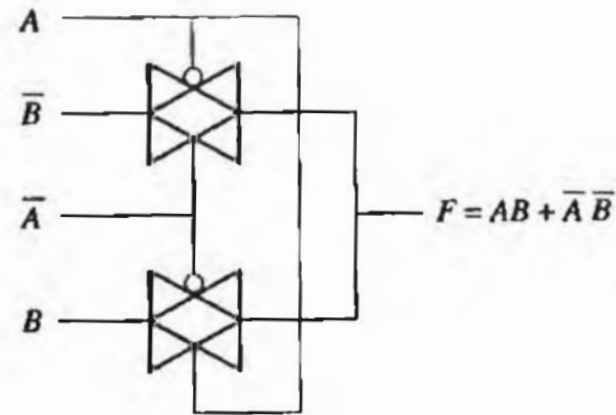


Multiplexer, XOR and XNOR

- 8 transistors for TG vs.
16 transistors for CMOS

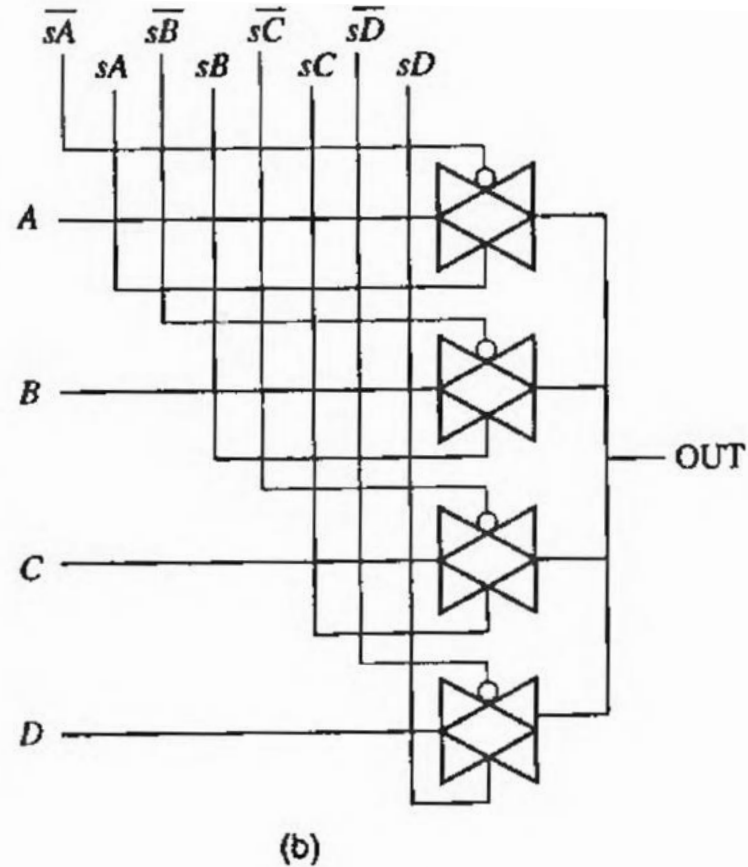
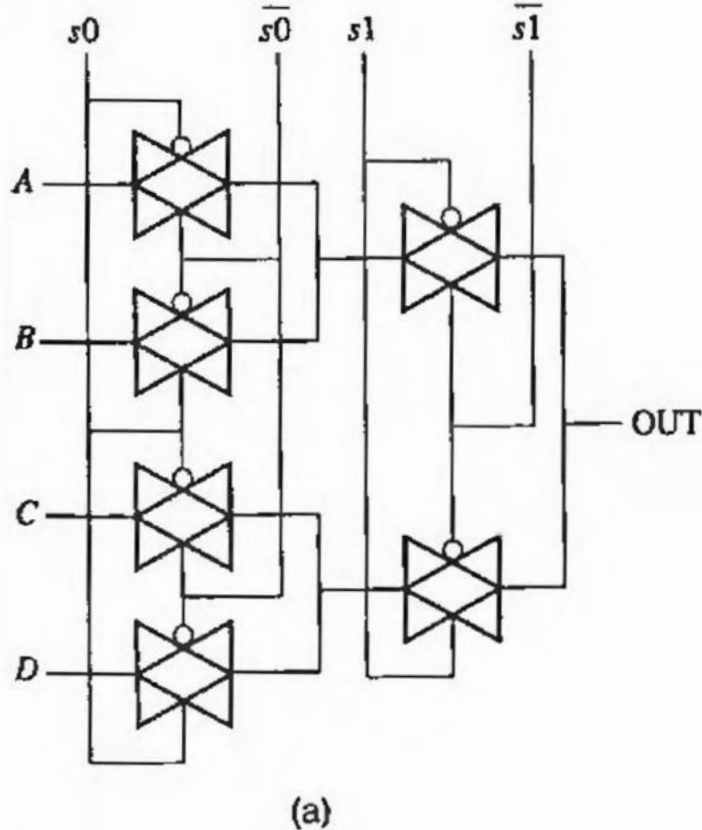


(a) XOR gate

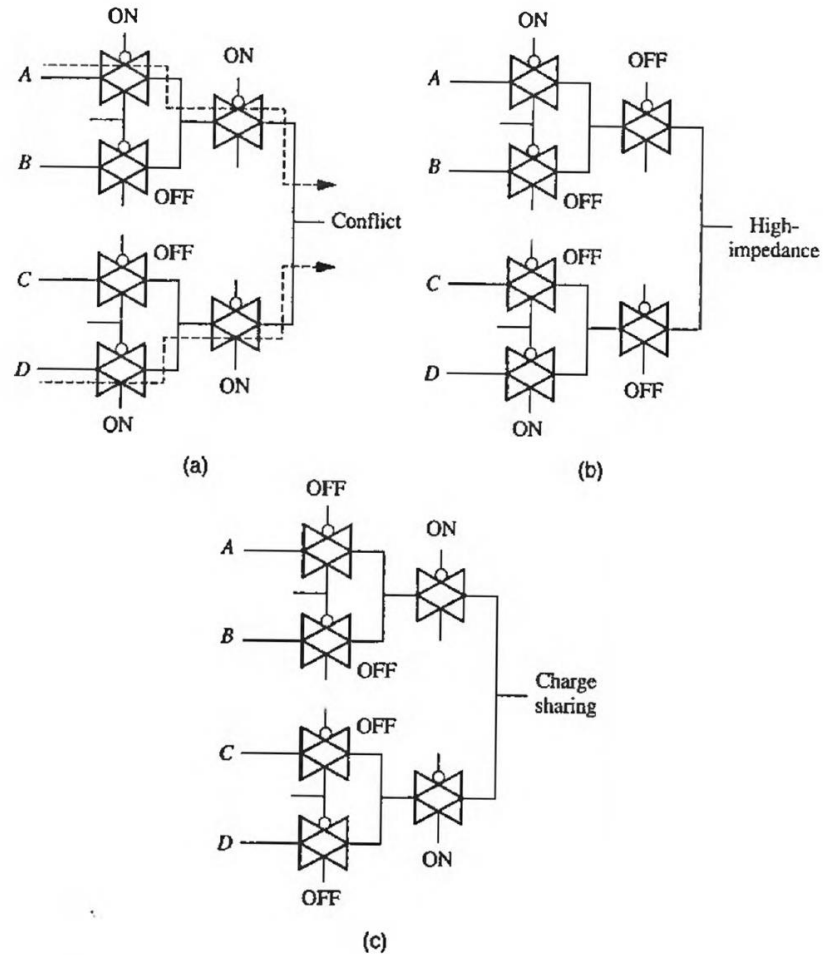


(b) XNOR gate

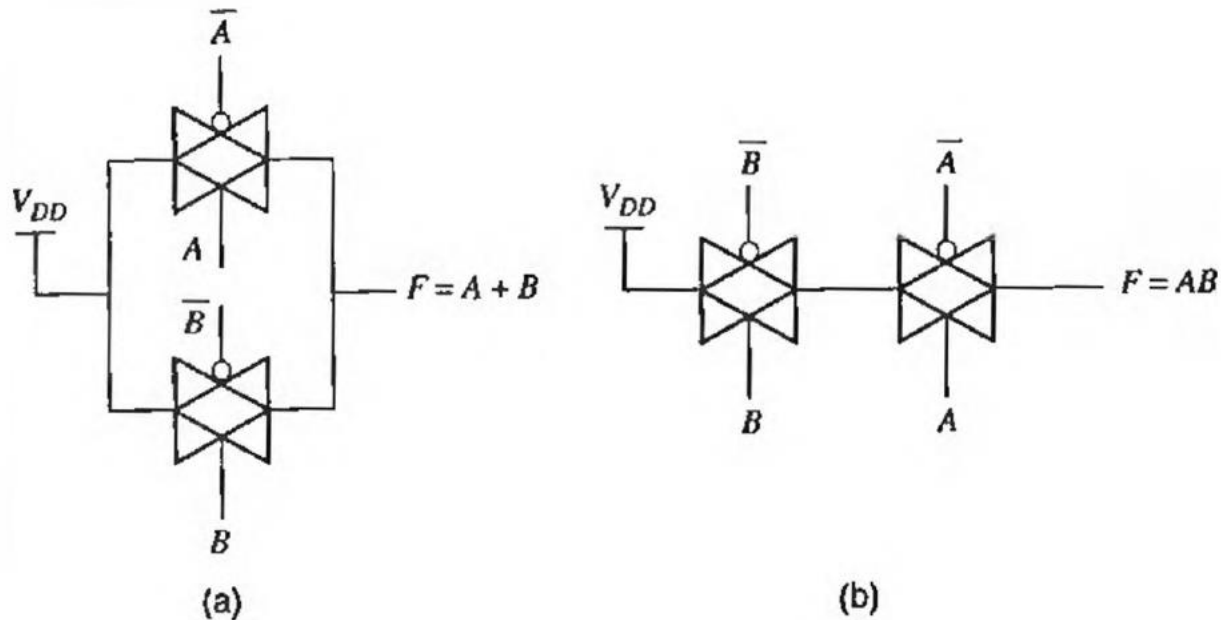
2- and Single-Level MUXs



Improper MUX Logic



OR and AND Functions



Ex. 7.3

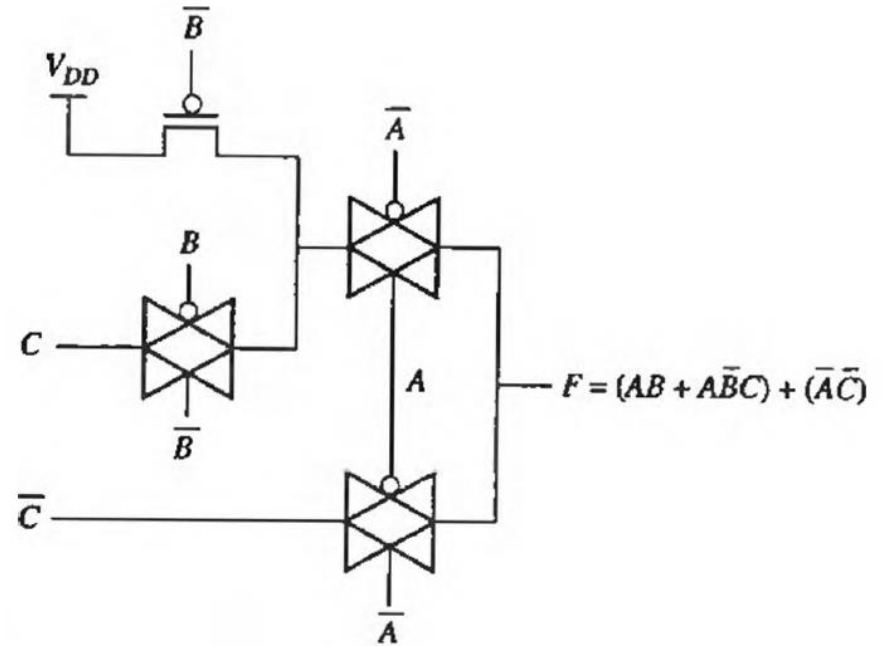
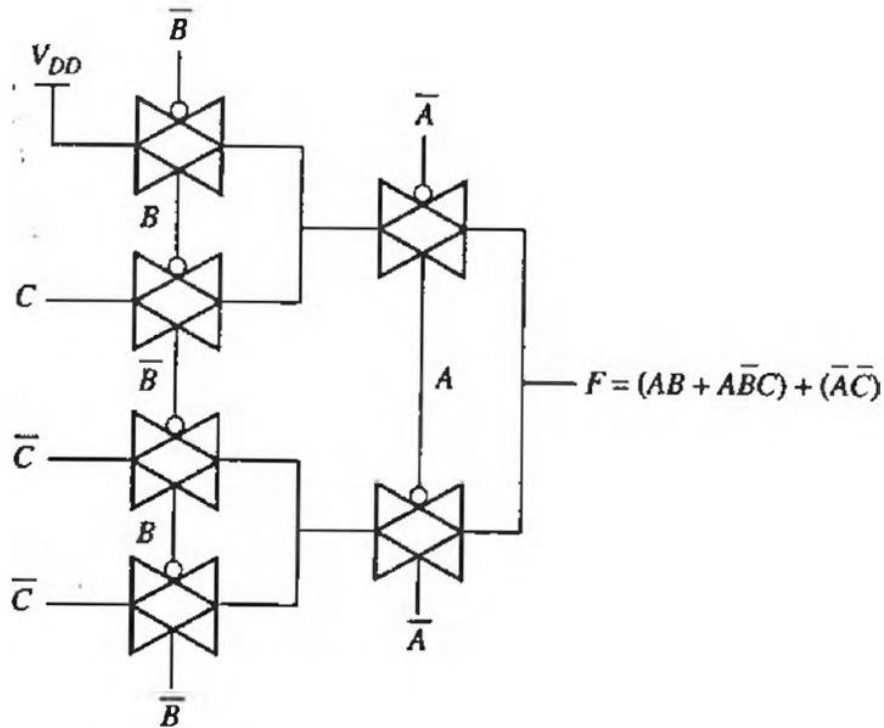
Implement the function $F = AB + \overline{A}\overline{B}C + \overline{A}C$ using transmission gates.

Solution:

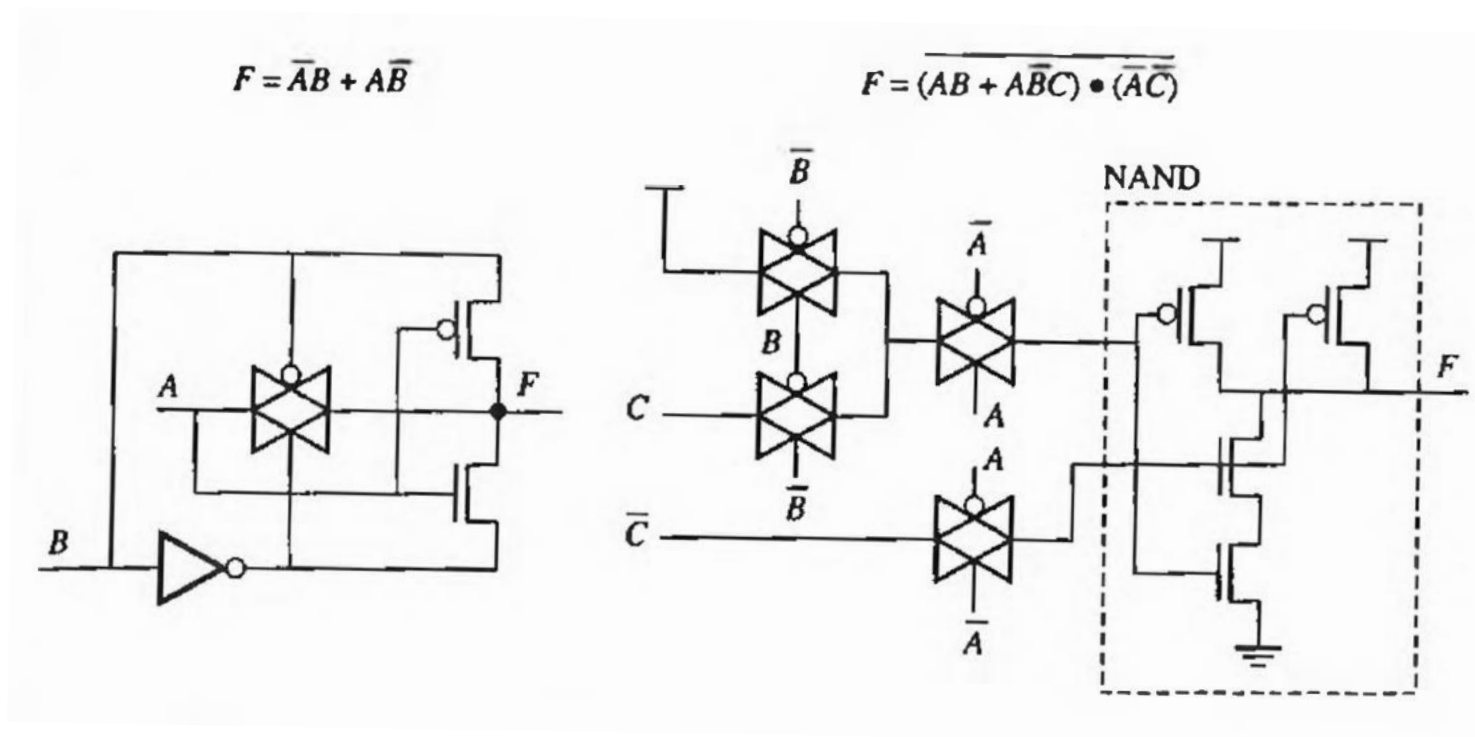
Identify A and B as control signals and build a truth table.

A	B	F
0	0	$\overline{C}$
0	1	$\overline{C}$
1	0	C
1	1	1

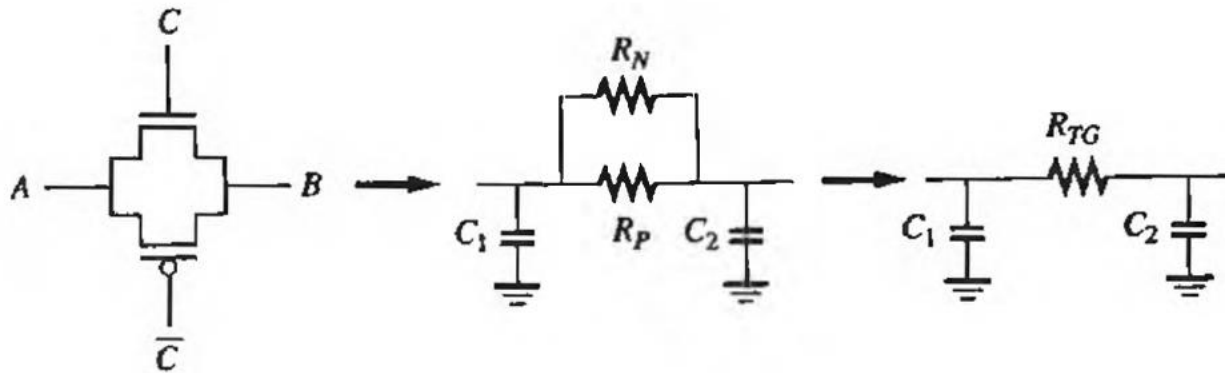
Ex. 7.3



TG and CMOS Gate Combinations



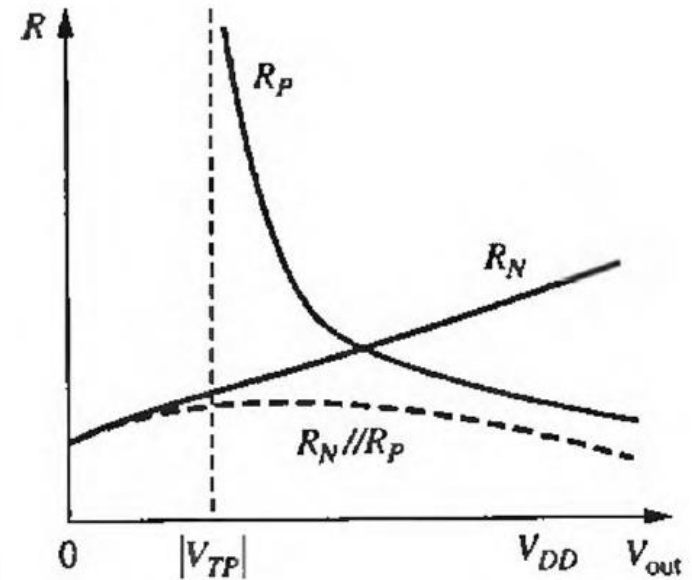
RC Model for CMOS TG



- $R_{TG} = R_N \parallel R_P$
- TG RC Model needs to be combined with RC models driver and load

TG On-Resistances

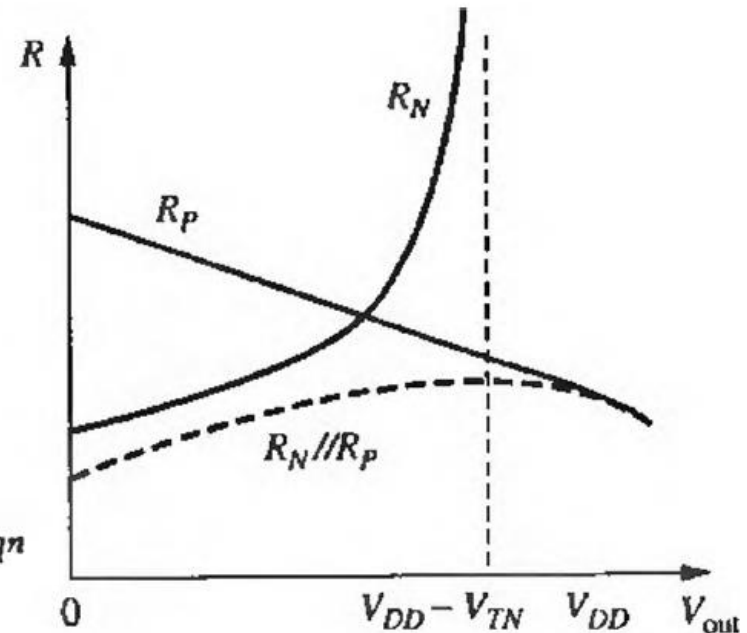
- When $V_{out} = V_{DD} \rightarrow 0$,
NMOS On and
 $R_N = R_{eqn} = 12.5\text{k}\Omega/\square$
PMOS initially on but
begins to shut off and
 $R_P = 2R_{eqp} = 4.8R_{eqn}$
- $R_{TG} = R_N // R_P = R_{eqn} // 2R_{eqp}$
 $\approx R_{eqn} // 4.8R_{eqn} = 0.83R_{eqn} \approx R_{eqn}$



(a) $V_{in} = 0\text{ V}$, $V_{out} = V_{DD} \rightarrow 0\text{ V}$

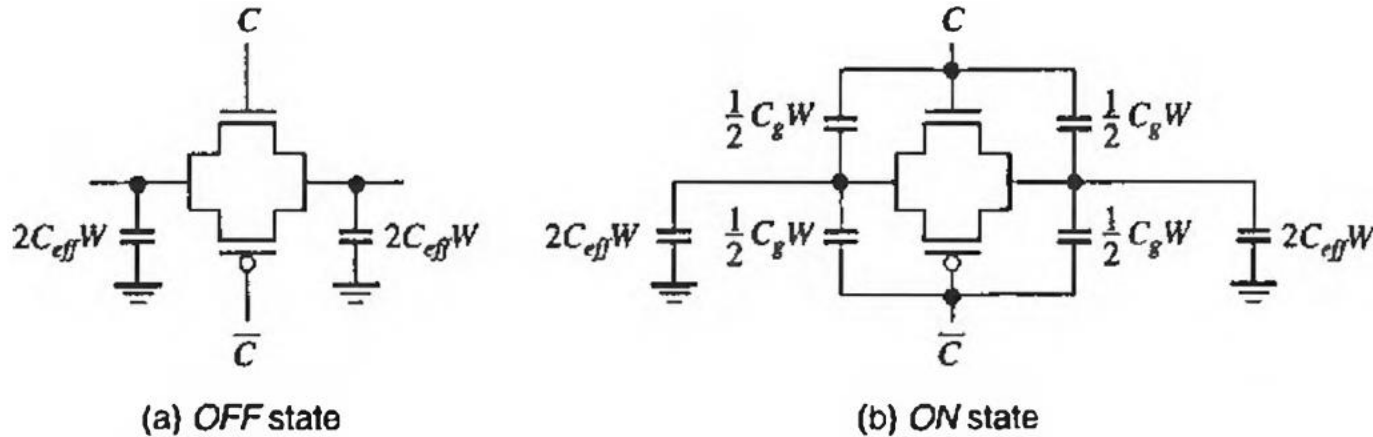
TG On-Resistances

- When $V_{out} = 0 \rightarrow V_{DD}$,
PMOS On and
 $R_P = R_{eqp} = 30k\Omega/\square$
NMOS initially on but
begins to shut off and
 $R_N = 2R_{eqn} = 25k\Omega/\square$
- $R_{TG} = R_N // R_P = 2R_{eqn} // R_{eqp}$
 $\approx 2R_{eqn} // 2.4R_{eqn} = 1.1R_{eqn} \approx R_{eqn}$



(b) $V_{in} = V_{DD}, V_{out} = 0 \text{ V} \rightarrow V_{DD}$

Transfer Gate Capacitances



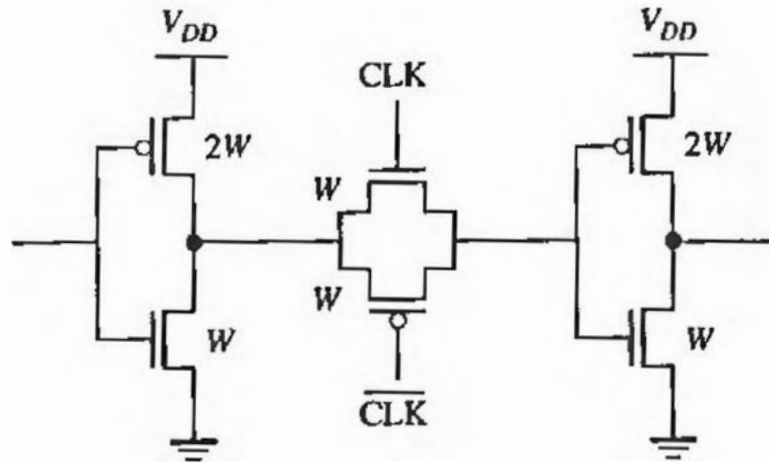
- OFF State: $C_{in} = C_{out} = C_{eff}(W_n + W_p)$

$$C_{in} = C_{out} = C_{eff}2W$$

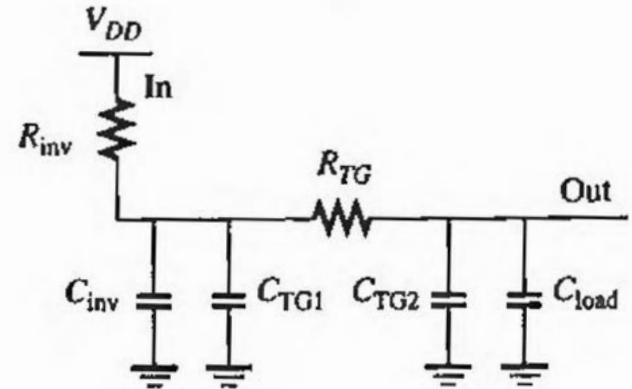
- ON State: $C_{in} = C_{out} = C_{eff}(W_n + W_p) + \frac{1}{2}(C_gW_n + C_gW_p)$

$$C_{in} = C_{out} = C_{eff}2W + C_gW$$

Transmission Gate with D and L



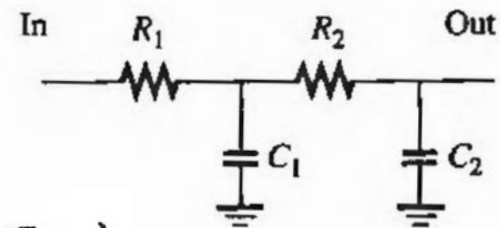
(a)



(b)

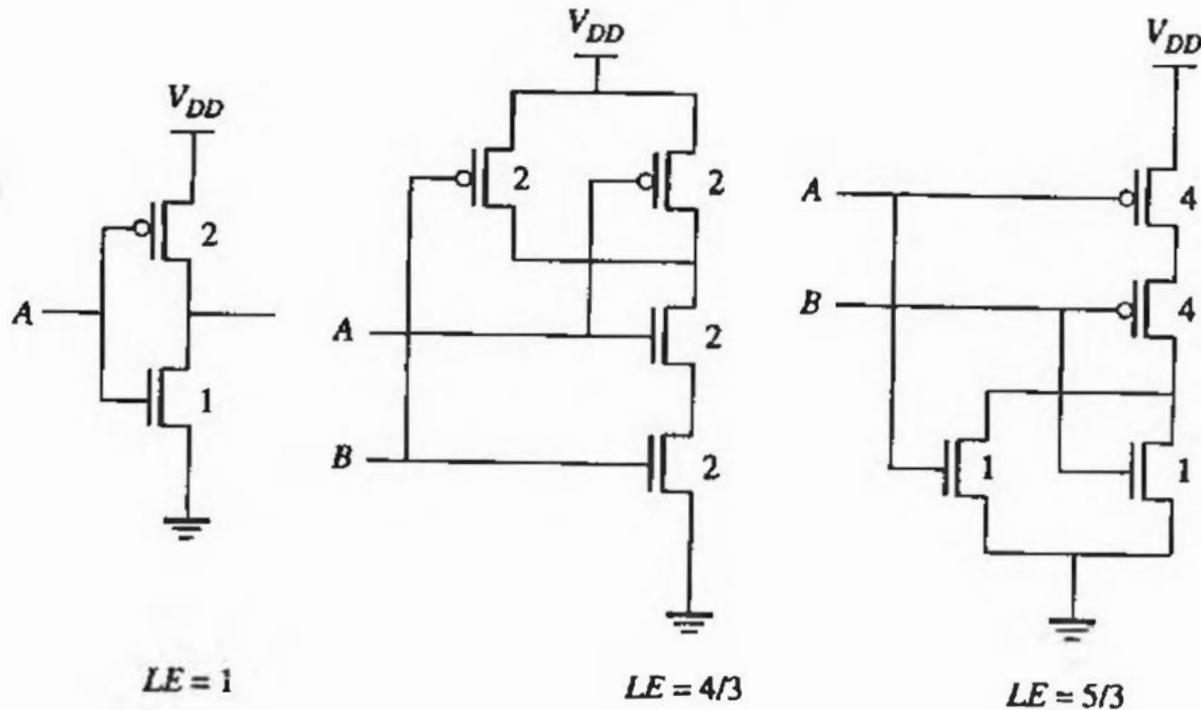
$$t_{\text{Elmore}} = R_1 C_1 + (R_1 + R_2) C_2$$

$$t_{\text{Elmore}} = R_{\text{inv}}(C_{\text{inv}} + C_{\text{TG1}}) + (R_{\text{inv}} + R_{\text{TG}})(C_{\text{TG2}} + C_{\text{load}})$$



(c)

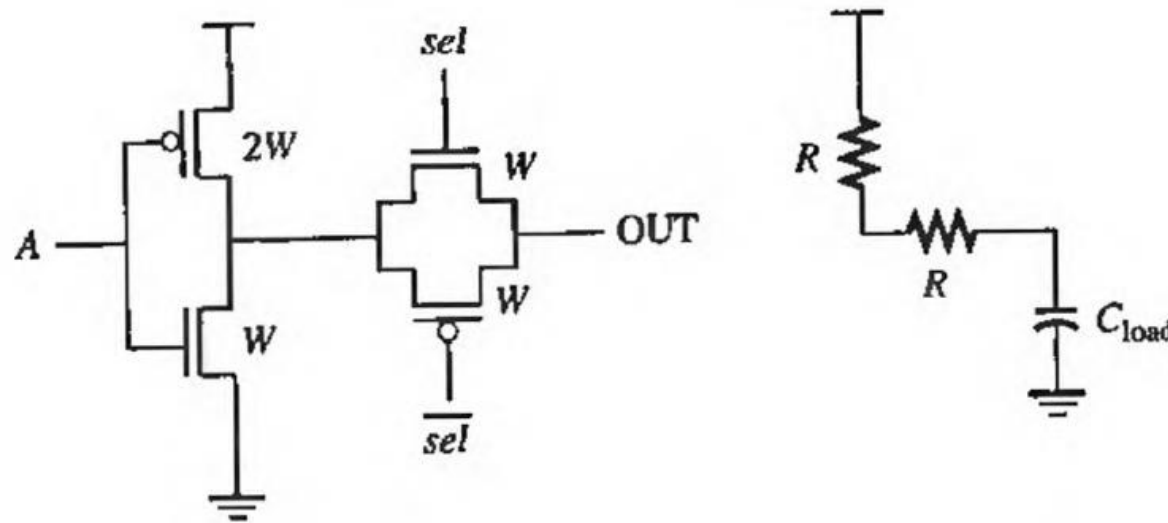
Logical Effort of Gates



For the NAND gate: $LE = \frac{(C_{in})_{nand}}{(C_{in})_{inv}} = \frac{2 + 2}{3} = \frac{4}{3}$

For the NOR gate: $LE = \frac{(C_{in})_{nor}}{(C_{in})_{inv}} = \frac{4 + 1}{3} = \frac{5}{3}$

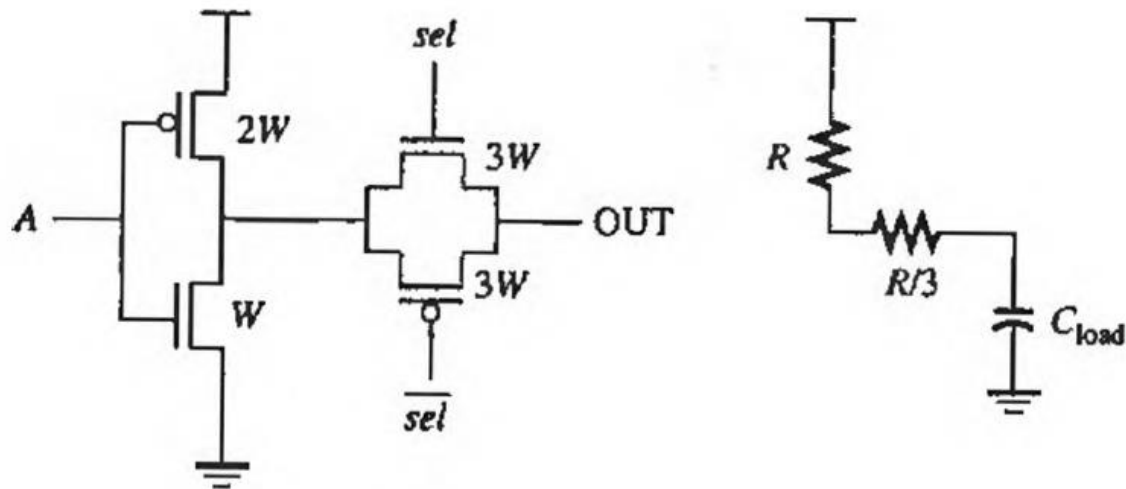
Logical Effort with CMOS TGs



$$\text{LE input } A = \frac{3W(2R)}{3WR} = 2$$

$$\text{LE input } sel = \frac{W(2R)}{3WR} = 2/3$$

LE Computation with 3X TG



$$\text{LE input } A = \frac{3W(4R/3)}{3WR} = 4/3$$

$$\text{LE input } sel = \frac{3W(4R/3)}{3WR} = 4/3$$